Predicting Credit Card Risk with Machine Learning Models

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Why Predict Credit Default?

Traditional rule-based systems are subjective and can't capture complex patterns

Goal: build a classifier that catches defaulters without flagging everyone

Only 1.7% of applicants default, extreme class imbalance makes this hard

Credit default costs financial institutions billions annually

Kaggle Credit Card Approval Prediction Dataset

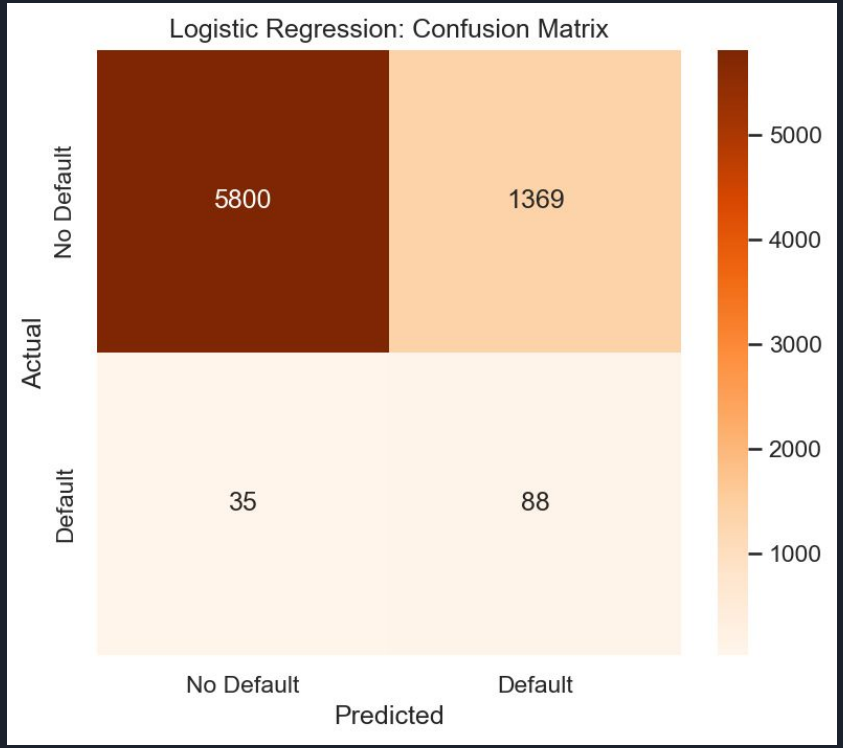
- application_record.csv: 438,557 records, 18 attributes (demographics, income, employment, housing)
- credit_record.csv: 1,048,575 monthly payment status records
- One-to-many merge on applicant ID → aggregated to 36,457 usable records
- Target: default = 1 if ever 60+ days overdue (STATUS 2/3/4/5); 616 defaulters vs 35,841 non-defaulters

Preprocessing Pipeline

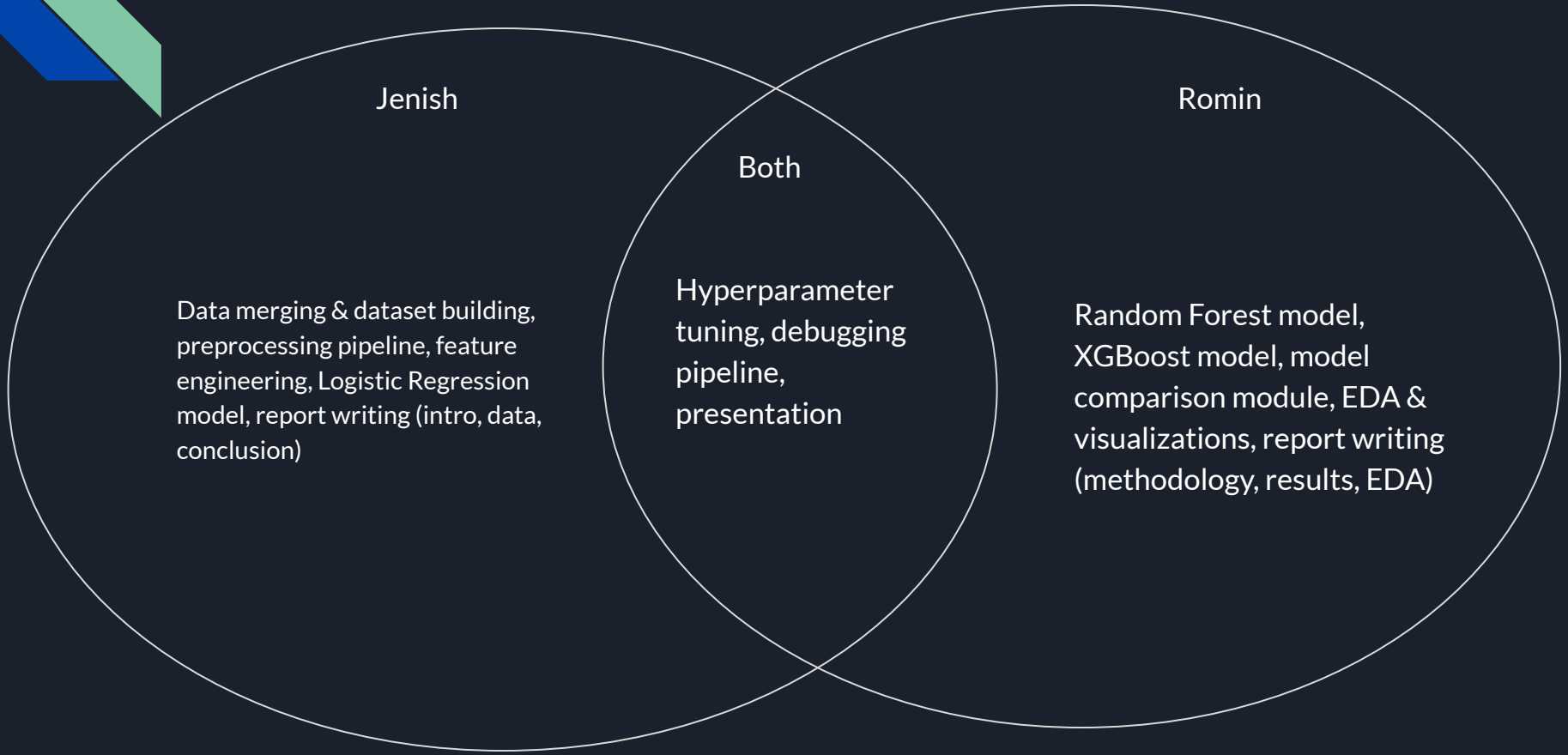
- Missing values: OCCUPATION_TYPE had 134K nulls — filled categorical with mode, numerical with median
- Outliers: IQR-based winsorization (capped at 1.5 x IQR boundaries)
- Encoding: Label encoding for binary columns, one-hot for multi-class (8 categorical features)
- Feature engineering: Created age_years, years_employed, income_per_member, income_per_child
- Scaling & split: StandardScaler on numericals, 80/20 stratified split → 29,165 train / 7,292 test

Logistic Regression: Baseline Model

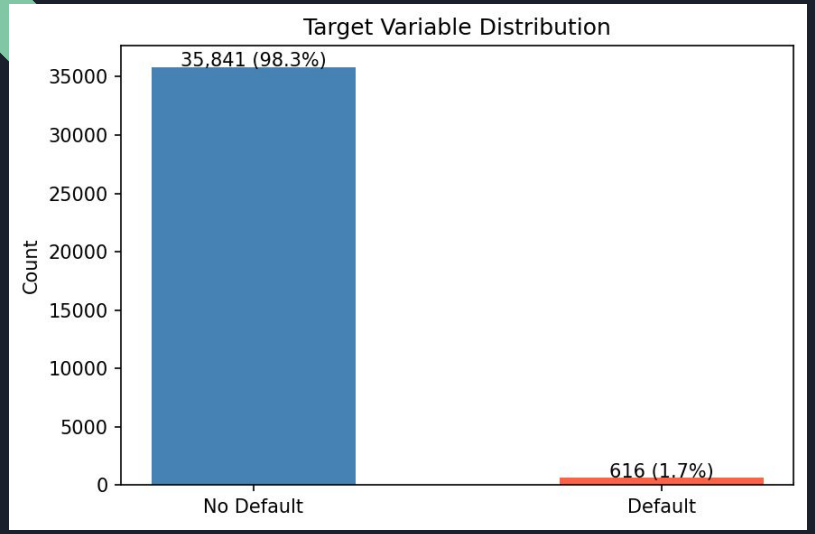
- SMOTE oversampling + GridSearchCV (5-fold CV)
- Searched: C = [0.001, 0.01, 0.1, 1, 10, 100], penalty = L1/L2, solver = liblinear/saga
- Best params: C=1, L1 penalty, balanced class weight
- Results: 80.8% accuracy, 72% recall, 0.81 ROC AUC
- Trade-off: catches most defaulters but with many false positives (6% precision)

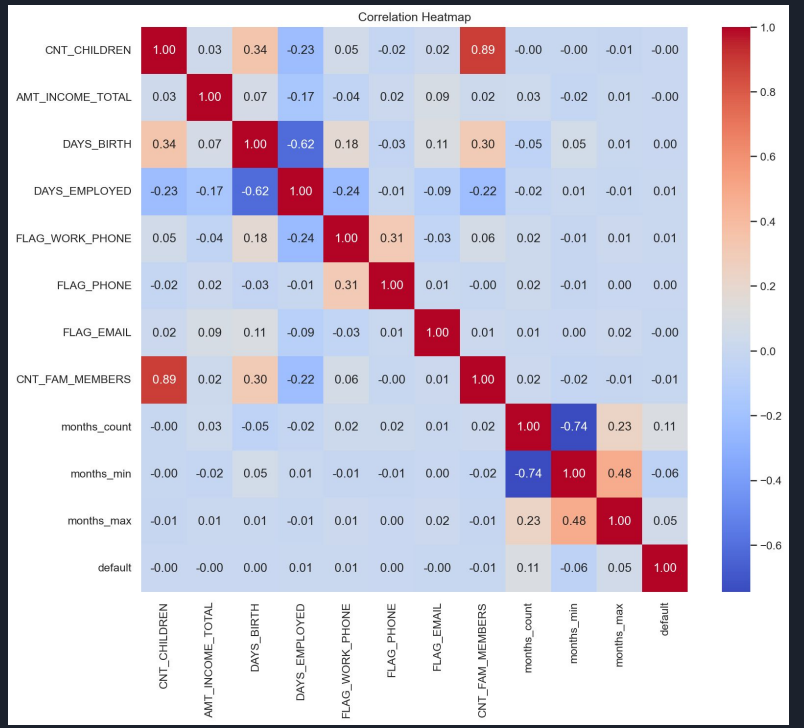


Work Distribution



Exploratory Data Analysis

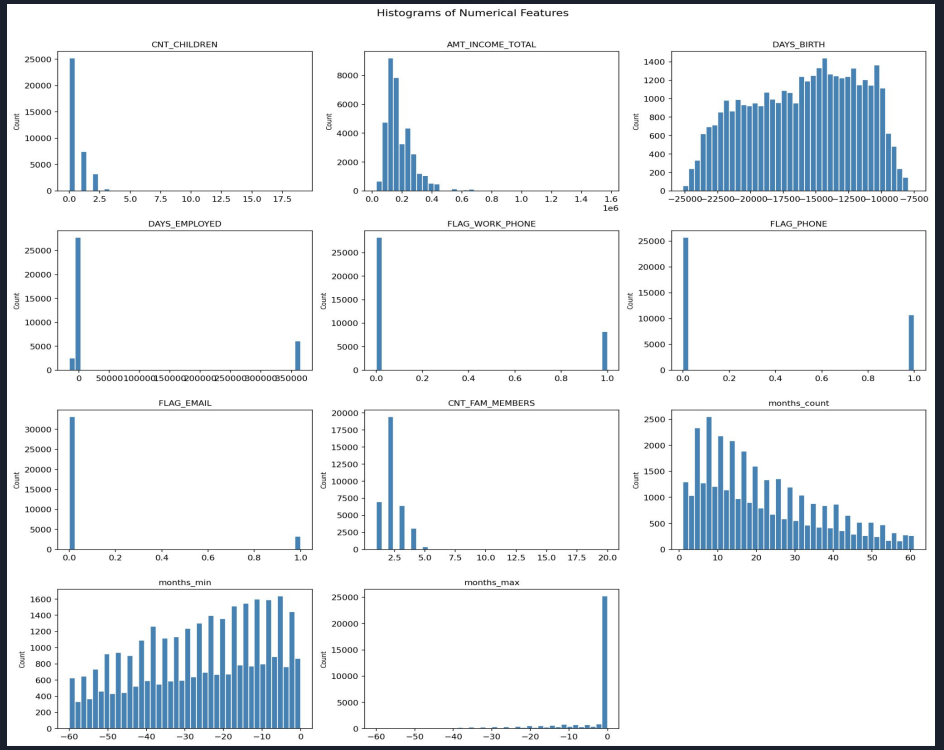




- Target is heavily imbalanced: 98.3% non-default vs 1.7% default (616 out of 36,457)
- OCCUPATION_TYPE has 134,203 missing values (~31% of records) — largest gap in the data
- Income distribution is right-skewed; most applicants earn under 380K annually
- Payment status "C" (paid off) dominates credit records — most applicants have healthy histories
- Correlation analysis: status_2 through status_5 counts are most predictive of default

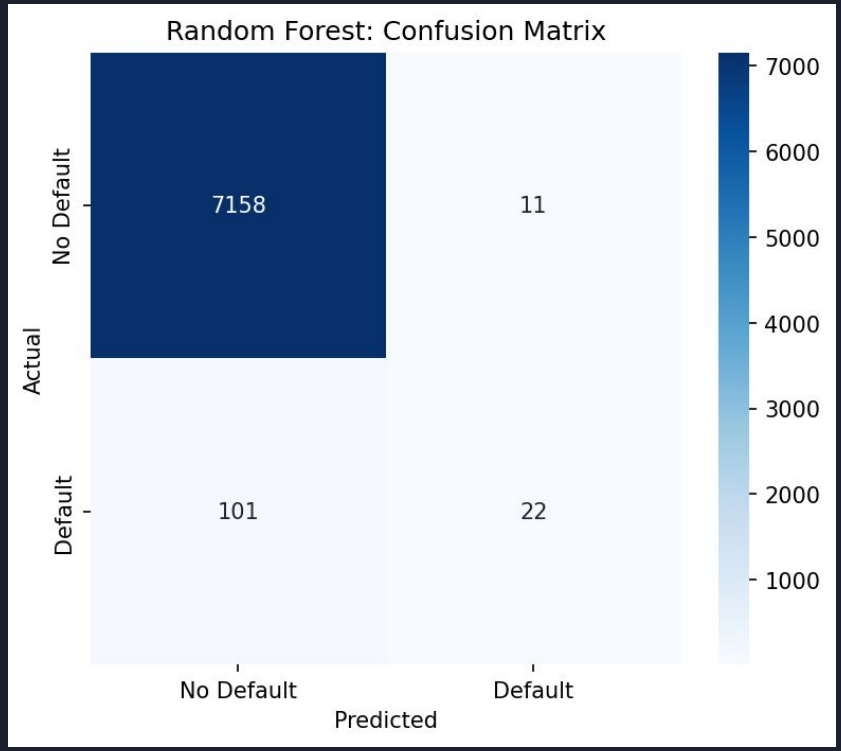
Feature Engineering

- Converted DAYS_BIRTH → age_years, DAYS_EMPLOYED → years_employed
- Created income_per_member (income / family size) and income_per_child
- Aggregated credit history: status frequency counts + months on record stats
- Removed low-variance features (threshold < 0.01) and highly correlated pairs (r > 0.9)
- Ranked features by mutual information — top predictors: status counts particularly status_0 and status_C and months_count

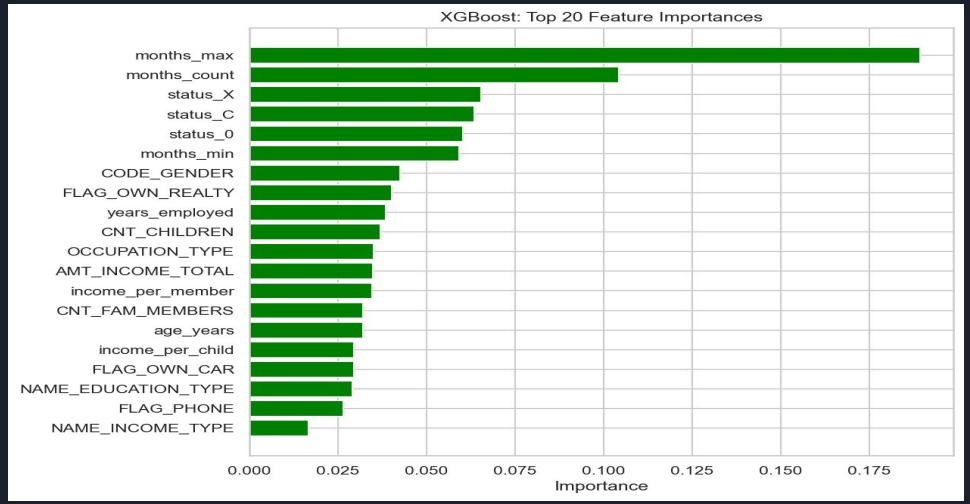


Random Forest

- SMOTE oversampling + balanced class weight in an imblearn pipeline
- RandomizedSearchCV (15 iterations, 5-fold CV, scoring = F1)
- Best params: 200 trees, max_depth=30, max_features=log2
- Results: 98.5% accuracy, 66.7% precision, 17.9% recall, 0.88 ROC AUC
- High precision but conservative — misses many defaulters to avoid false alarms



XGBoost

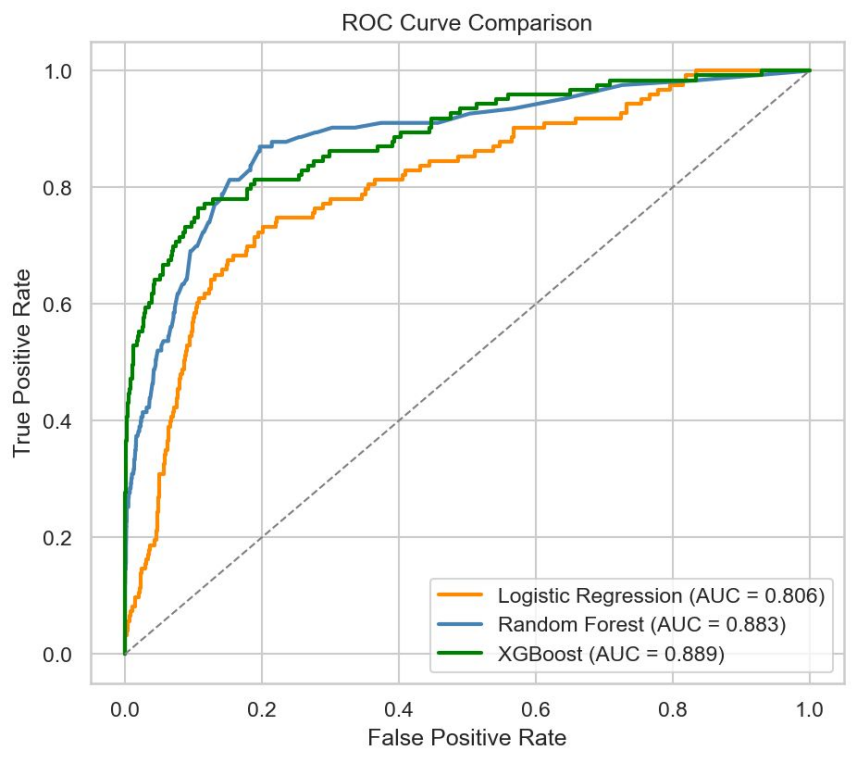


- Handles imbalance via scale_pos_weight (ratio ~58:1, auto-calculated from data)
- RandomizedSearchCV (20 iterations, 5-fold CV, scoring = F1)
- Best params: 300 trees, max_depth=5, learning_rate=0.2, subsample=1.0
- Results: 98.2% accuracy, 49.6% precision, 49.6% recall, 0.89 ROC AUC (highest)
- Best precision-recall balance among all three models

Model Comparison

Model	Accuracy	Precision	Recall	F1	ROC AUC
Logistic Regression	80.8%	6.0%	71.5%	0.11	0.81
Random Forest	98.5%	66.7%	17.9%	0.28	0.89
XGBoost	**98.3%**	**49.6%**	**49.6%**	**0.50**	**0.89**

ROC Comparison



Conclusion & Future Work

- XGBoost is the best model: highest ROC AUC (0.89), best F1 (0.50), 98.3% accuracy
- Ensemble methods significantly outperform logistic regression for imbalanced credit data
- Class imbalance handling (SMOTE, class weights, scale_pos_weight) is essential — not optional
- Future work: deep learning on sequential payment data, stacking all 3 models, fairness auditing across demographic groups